

SUPERVISED MACHINE LEARNING
SGD Classification

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2) SGD Classifier of Linear Model

The **SGD Classifier** is an algorithm of classification that is a part of the linear model module.

```
1 import numpy as np
2 import pandas as pd
3 import matplotlib.pyplot as plt
4
5 from sklearn.datasets import load_iris
6 from sklearn.linear_model import SGDClassifier
7 from sklearn.metrics import r2_score, mean_squared_error, mean_absolute_error
```

Here, we try to classify a type of flower though the **iris** dataset.

```
1 iris = load_iris()
2 y = iris.target
3 X = iris.data
4
5 modele = SGDClassifier(random_state = 0)
6 modele.fit(X,y)
7
8 y_pred = modele.predict(X)
9 R2_sc = r2_score(y, y_pred)
10 R2 = modele.score(X, y)
11
12 mse = mean_squared_error(y, y_pred)
13 rmse = mean_squared_error(y, y_pred, squared = False)
14 mae = mean_absolute_error(y, y_pred)
15
16 print(f"\nScore (Error in the prediction): {R2}")
17 print(f"Score (Error in the datasets): {R2_sc}\n")
18
19 print(f"MSE = {mse}")
20 print(f"RMSE = {rmse}")
21 print(f"MAE = {mae}")
22
23 plt.scatter(X[:,0], y, c = 'b', label = 'Real datasets')
24 plt.scatter(X[:,0], y_pred, c = 'r', label = 'Prediciton datasets')
25 plt.legend()
26 plt.show()
```

```
plt.close()
```

Score (Error in the prediction): 0.9666666666666667

Score (Error in the datasets): 0.95

MSE = 0.03333333333333333

RMSE = 0.18257418583505536

MAE = 0.03333333333333333

